

Wall-crossing residues, Poincaré sums, and the DJ2008 picture

Working notes companion to `dr_b_periods_and_Lfunctions.tex`

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Purpose

These are working musings on the structural relationship between the wall-crossing residue formula of the main note (`dr_b_periods_and_Lfunctions.tex`, Theorem 1.3, the residue formula $\mathcal{R}_{\tau_p}(f, s)$ there) and the Poincaré-sum-of-residues constructions of Hardy–Ramanujan 1919 [1] and Duke–Jenkins 2008 [2]. The aim is to identify the right structural slot in the literature for our cumulative wall-crossing residue, and to record open questions about convergence and modular identification of the resulting object. Nothing here is proved at the level required for the main paper; this is a scratchpad.

1 Recap: the single-crossing residue

For a meromorphic modular form f of weight k on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ and an admissible contour pair $(\gamma_S(\tau_0), \gamma_T(\tau_0))$ avoiding the Γ -orbit of $f^{-1}(\infty)$, the finite-contour L -functional

$$L^*(f, s; \gamma_S, \gamma_T) = i^{-s} \left[\int_{\gamma_S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (1)$$

with Hurwitz-zeta T -kernel $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$ depends on the homotopy class of the contour pair only. A single crossing of an interior pole τ_p of f contributes

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \mathrm{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (2)$$

with $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ the winding contributions of γ_S, γ_T across τ_p .

2 Residue transport along Γ -orbit

If f is meromorphic modular of weight k , then for $\gamma_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ acting by $\gamma_0\tau = (a\tau + b)/(c\tau + d)$, the residue at the orbit-translated pole satisfies

$$\mathrm{Res}(f, \gamma_0\tau_p) = (c\tau_p + d)^{k-2} \mathrm{Res}(f, \tau_p). \quad (3)$$

The cocycle factor $(c\tau_p + d)^{k-2}$ comes from combining the weight- k transformation $f(\gamma_0\sigma) = (c\sigma + d)^k f(\sigma)$ with the Jacobian $(\gamma_0\sigma - \gamma_0\tau_p) = (\sigma - \tau_p)/((c\sigma + d)(c\tau_p + d))$ in the residue limit. This is the standard residue transport on $\mathbb{H}$ under Γ .

3 Cumulative wall-crossing as a Poincaré-style sum

Consider f with poles at exactly one Γ -orbit $\Gamma \cdot \tau_p \subset \mathbb{H}$. A continuous deformation of a contour pair from one homotopy class enclosing no orbit-points to one enclosing every orbit-point produces a cumulative wall-crossing

$$\Delta L^*(f, s) = -2\pi i i^{-s} \text{Res}(f, \tau_p) \sum_{\gamma_0 \in \Gamma_{\tau_p} \setminus \Gamma} (c\tau_p + d)^{k-2} \left[\sigma^S(\gamma_0) (\gamma_0 \tau_p)^{s-1} + \sigma^T(\gamma_0) \tilde{k}_T(\gamma_0 \tau_p, s) \right] \quad (4)$$

where Γ_{τ_p} is the stabiliser of τ_p (trivial except at elliptic points i and ρ). Pulling out the kernel structure, define

$$K_f(\tau_p, s) := \sum_{\gamma_0 \in \Gamma_{\tau_p} \setminus \Gamma} (c\tau_p + d)^{k-2} \left[\sigma^S(\gamma_0) (\gamma_0 \tau_p)^{s-1} + \sigma^T(\gamma_0) \tilde{k}_T(\gamma_0 \tau_p, s) \right], \quad (5)$$

formally. The single-orbit cumulative wall-crossing is $\Delta L^* = -2\pi i i^{-s} \text{Res}(f, \tau_p) K_f(\tau_p, s)$, plus a finite-deformation correction for any portion of the orbit not swept by the contour deformation.

3.1 Transformation properties of $K_f(\tau_p, s)$

Under $\tau_p \mapsto \gamma_1 \tau_p$ for $\gamma_1 \in \Gamma$, the orbit sum reindexes via $\gamma_0 \mapsto \gamma_0 \gamma_1^{-1}$, and the cocycle factor $(c\tau_p + d)^{k-2}$ transforms by the weight- $(k-2)$ cocycle. In particular,

$$K_f(\gamma_1 \tau_p, s) = (c_1 \tau_p + d_1)^k K_f(\tau_p, s) \quad (6)$$

formally, i.e., K_f is weight- k in τ_p , with s acting as a spectator parameter.

3.2 Comparison with HR1919 Theorem 2

Hardy–Ramanujan [1, Thm. 2] prove that a modular function ϕ (their convention: weight n under $\phi((c+d\tau)/(a+b\tau)) = (a+b\tau)^n \phi(\tau)$, $ad-bc=1$) with a single simple pole at $\tau = \alpha$ and residue A admits the representation

$$f(q) = 2\pi i A \sum \frac{1}{(a+b\alpha)^{n+2}} \frac{\mathbf{q}^2}{q^2 - \mathbf{q}^2}, \quad \mathbf{q} = \exp\left(\frac{c+d\alpha}{a+b\alpha} \pi i\right), \quad (7)$$

summed over pairs (a, b) giving distinct $\mathbf{q}$. Each summand is the residue at one orbit-point, transported by the modular cocycle $(a+b\alpha)^{-(n+2)}$ and weighted by an explicit q -kernel.

Structural match. HR1919’s orbit-of-residues structure is identical to ours: sum over a Γ -orbit, cocycle weighting that transports the base residue, times a kernel evaluated at the orbit-point. Conventions differ (HR’s $(a+b\alpha)^{-(n+2)}$ versus our $(c\tau_p + d)^{k-2}$): HR work with a non-standard slash-action where the cocycle appears in the inverse, and their n corresponds to our $k-2$ in some weight-conversion sense. The structural role of the cocycle is identical.

Structural mismatch. HR’s ϕ is a fixed meromorphic modular function; the kernel $\mathbf{q}^2/(q^2 - \mathbf{q}^2)$ is fixed. Our cumulative wall-crossing $K_f(\tau_p, s)$ is a one-parameter analytic family in s . At fixed s , the analog of HR’s identity would be

$$\phi_{f,s}(\tau_p) = \text{Res}(f, \tau_p) K_f(\tau_p, s) + (\text{regular part}) \quad (8)$$

where $\phi_{f,s}$ is the conjectural weight- k object on $\mathbb{H} \setminus \Gamma \cdot \tau_p$ whose residue at τ_p is $\text{Res}(f, \tau_p)$, and whose principal part at each orbit-point is determined by the cocycle factor in (5).

3.3 Comparison with DJ2008 Theorem 2

Duke–Jenkins [2, Thm. 2], working with weakly holomorphic forms $\mathcal{M}_k$ at level one with the canonical basis $f_{k,m}(\tau) = q^{-m} + O(q^{\ell+1})$ for $m \geq -\ell = -\lfloor k/12 \rfloor + \text{correction}$, prove that

$$\sum_{m \geq -\ell} f_{k,m}(z) q^m = \frac{f_k(z) f_{2-k}(\tau)}{j(\tau) - j(z)}, \quad (9)$$

where $f_k = \Delta^\ell E_{k'}$ and $q = e^{2\pi i \tau}$. The RHS has singularities exactly on the Γ -orbit $\Gamma \cdot z$. At $\tau = \gamma_0 z$ for $\gamma_0 \in \Gamma$,

$$\text{Res}_{\tau=\gamma_0 z} \left[\frac{f_k(z) f_{2-k}(\tau)}{j(\tau) - j(z)} \right] = \frac{f_k(z) f_{2-k}(z)}{j'(z)} \cdot (cz + d)^{-k}, \quad (10)$$

using $f_{2-k}(\gamma_0 z) = (cz + d)^{2-k} f_{2-k}(z)$ and $j'(\gamma_0 z) = (cz + d)^2 j'(z)$. The residues along $\Gamma \cdot z$ are weighted by the weight- k modular cocycle $(cz + d)^{-k}$, with all other γ_0 -dependence factored out.

Structural match. DJ2008’s residue weighting along $\Gamma \cdot z$ is exactly the weight- k Poincaré cocycle structure. Our $K_f(\tau_p, s)$ has the analogous structure with $(c\tau_p + d)^{k-2}$ (our cocycle is on the residue of f at τ_p , theirs is on the kernel $1/(j(\tau) - j(z))$ at z — same role).

Crucial methodological alignment. DJ2008 emphasise that for $\ell \neq 0$ (i.e. for genuinely weakly holomorphic forms with cusp poles), the classical Petersson Poincaré series $P_{k,m}$ *do not* represent the basis elements $f_{k,m}$:

“In order to prove Theorem 1 in general, we will avoid the use of Poincaré series and Hecke operators, since the relations (11) and (13) need not hold when $\ell \neq 0$. Instead, we derive an integral formula for $f_{k,m}$, for which approximation by residues leads to Theorem 1.”

Their proof method is exactly the residue-approximation logic that governs our wall-crossing: the cumulative residue sum reconstructs the function. This is the right structural slot for our wall-crossing residues — not classical convergent Petersson Poincaré series, but DJ2008-style residue-integral reconstruction.

4 The $\tau_0 = i$ specialisation: explicit orbit sums

At the elliptic-fixed-point basepoint $\tau_0 = i$ used throughout the main note, $S \cdot i = -1/i = i$, so the S -segment $[-1/i, i] = \{i\}$ collapses to a point. Consequently $\sigma^S(\gamma_0) = 0$ for every $\gamma_0 \in \Gamma$ — there is no S -segment that can cross a pole — and the wall-crossing structure reduces to that of the T -segment alone. This section makes the orbit-sum structure explicit in this specialisation, for the L -integral and the period-integral separately, and discusses the $k < 0$ regime where the orbit sum converges naively as in HR1919.

4.1 L -integral at $\tau_0 = i$: T -segment-only orbit sum

At $\tau_0 = i$ the single-crossing residue (2) reduces to

$$\mathcal{R}_{\tau_p}(f, s) \Big|_{\tau_0=i} = -2\pi i i^{-s} \sigma_{\tau_p}^T \text{Res}(f, \tau_p) \widetilde{k}_T(\tau_p, s). \quad (11)$$

The cumulative wall-crossing as the T -segment $[i-1, i]$ is deformed continuously to enclose every point in the Γ -orbit $\Gamma \cdot \tau_p$ is therefore

$$\Delta L^*(f, s)|_{\tau_0=i} = -2\pi i i^{-s} \text{Res}(f, \tau_p) K_f^T(\tau_p, s), \quad (12)$$

where

$$K_f^T(\tau_p, s) := \sum_{\gamma_0 \in \Gamma_{\tau_p} \setminus \Gamma} (c\tau_p + d)^{k-2} \sigma^T(\gamma_0) \tilde{k}_T(\gamma_0 \tau_p, s). \quad (13)$$

This is the T -only specialisation of $K_f(\tau_p, s)$ in (5): a one-parameter analytic family in s , formally weight- k in τ_p , written as a single-kernel Poincaré-style sum on the Γ -orbit of τ_p .

At integer $s = j \in \{2, \dots, k-2\}$. The Hurwitz-zeta kernel collapses to a polynomial in τ : $\tilde{k}_T(\tau, j)$ is a $\mathbb{Q}$ -polynomial built from Bernoulli polynomials $B_m(\tau+1)$. Equivalently, in the $\omega^\pm$ -normalised period kernel of the main note's Theorem 1.1, $\tilde{k}_T(\tau, j) = (-1)^{j-1} \binom{k-2}{j-1} k_T(j, \tau)$, where $k_T(j, \tau)$ is the polynomial T -kernel. The orbit sum at integer $s = j$ becomes

$$K_f^T(\tau_p, j) = (-1)^{j-1} \binom{k-2}{j-1} \sum_{\gamma_0} (c\tau_p + d)^{k-2} \sigma^T(\gamma_0) k_T(j, \gamma_0 \tau_p), \quad (14)$$

a Poincaré sum of the polynomial kernels $k_T(j, \cdot)$ along the orbit.

Explicit residues at the first few orbit points ($k = 12$)

To make the orbit sum (13) concrete, fix $k = 12$ (the running case of the main note) and let $\tau_p \in \mathbb{H}$ be a generic interior simple pole of f , not on the Γ -orbit of i or $\rho = e^{i\pi/3}$ (so $\Gamma_{\tau_p} = \{\pm I\}$ and the orbit $\Gamma \cdot \tau_p$ is genuinely $\text{PSL}_2(\mathbb{Z})$ -sized). For each coset representative $\gamma_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_{\tau_p} \setminus \Gamma$, the single-crossing residue from (11) applied at the orbit-point $\gamma_0 \tau_p$ is

$$\mathcal{R}_{\gamma_0 \tau_p}(f, s)|_{\tau_0=i} = -2\pi i i^{-s} \sigma^T(\gamma_0) (c\tau_p + d)^{10} \text{Res}(f, \tau_p) \tilde{k}_T(\gamma_0 \tau_p, s), \quad (15)$$

using the residue-transport identity $\text{Res}(f, \gamma_0 \tau_p) = (c\tau_p + d)^{k-2} \text{Res}(f, \tau_p)$ of §2 and the cocycle factor $(c\tau_p + d)^{k-2} = (c\tau_p + d)^{10}$ at $k = 12$.

The first few cosets, ordered by minimum $|c| + |d|$:

$$\gamma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad \gamma_0 \tau_p = \tau_p, \text{ cocycle factor } (0 \cdot \tau_p + 1)^{10} = 1.$$

$$\mathcal{R}_{\tau_p}|_{\tau_0=i} = -2\pi i i^{-s} \sigma^T(I) \text{Res}(f, \tau_p) \tilde{k}_T(\tau_p, s). \quad (16)$$

$$\gamma_0 = T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \quad \gamma_0 \tau_p = \tau_p + 1, \text{ cocycle factor } 1.$$

$$\mathcal{R}_{\tau_p+1}|_{\tau_0=i} = -2\pi i i^{-s} \sigma^T(T) \text{Res}(f, \tau_p) \tilde{k}_T(\tau_p + 1, s). \quad (17)$$

Using the shift-difference identity satisfied by $\tilde{k}_T$ (main note, §4.1 derivation of $\tilde{k}_T$): $\tilde{k}_T(\tau+1, s) = \tilde{k}_T(\tau, s) + (-\tau^{s-1} + e^{i\pi(s-1)} \tau^{11-s})$, (17) can be re-expressed entirely in terms of $k_T(\tau_p, s)$ plus an explicit closed-form correction.

$\gamma_0 = T^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. $\gamma_0 \tau_p = \tau_p - 1$, cocycle factor 1.

$$\mathcal{R}_{\tau_p-1}|_{\tau_0=i} = -2\pi i i^{-s} \sigma^T(T^{-1}) \text{Res}(f, \tau_p) \tilde{k}_T(\tau_p - 1, s). \quad (18)$$

Note that the T -segment $[i-1, i]$ at $\tau_0 = i$ has length one in the real direction. The $T^{\pm 1}$ translates of τ_p sit within one unit of τ_p in Re and at the same Im as τ_p , so for many physical contour deformations the $T^{\pm 1}$ -translates are encountered as immediate neighbours of τ_p .

$\gamma_0 = S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. $\gamma_0 \tau_p = -1/\tau_p$, cocycle factor $(1 \cdot \tau_p + 0)^{10} = \tau_p^{10}$.

$$\mathcal{R}_{-1/\tau_p}|_{\tau_0=i} = -2\pi i i^{-s} \sigma^T(S) \tau_p^{10} \text{Res}(f, \tau_p) \tilde{k}_T(-1/\tau_p, s). \quad (19)$$

For τ_p in the standard fundamental domain (so $|\tau_p| \geq 1$ and $\text{Im}(\tau_p) \geq \sqrt{3}/2$), the S -image $-1/\tau_p$ has $|-1/\tau_p| \leq 1$ and $\text{Im}(-1/\tau_p) = \text{Im}(\tau_p)/|\tau_p|^2 \leq \text{Im}(\tau_p)$. The cocycle factor τ_p^{10} is generically large (growing like $|\tau_p|^{10}$).

$\gamma_0 = ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$. $\gamma_0 \tau_p = -1/(\tau_p + 1)$, cocycle factor $(\tau_p + 1)^{10}$.

$$\mathcal{R}_{-1/(\tau_p+1)}|_{\tau_0=i} = -2\pi i i^{-s} \sigma^T(ST) (\tau_p + 1)^{10} \text{Res}(f, \tau_p) \tilde{k}_T(-1/(\tau_p + 1), s). \quad (20)$$

Running total. The cumulative wall-crossing residue after a deformation that encloses N orbit-points with winding numbers $\{\sigma^T(\gamma_0)\}_{\gamma_0 \in S_N}$ (where $S_N \subset \Gamma_{\tau_p} \setminus \Gamma$ is the swept-out subset) is the partial sum

$$\Delta L^*(f, s)|_{\tau_0=i, N\text{-fold}} = -2\pi i i^{-s} \text{Res}(f, \tau_p) \sum_{\gamma_0 \in S_N} (c\tau_p + d)^{10} \sigma^T(\gamma_0) \tilde{k}_T(\gamma_0 \tau_p, s). \quad (21)$$

The cumulative orbit sum $K_f^T(\tau_p, s)$ of (13) is the formal $N \rightarrow \infty$ limit (with S_N exhausting $\Gamma_{\tau_p} \setminus \Gamma$), modulo the convergence questions of §4.3: at $k = 12$ the cocycle factor $(c\tau_p + d)^{10}$ grows in $|c|, |d|$, so the infinite-orbit sum requires regularisation. Each finite-deformation partial sum (21) is, however, a well-defined finite quantity.

Integer- s specialisation in the critical strip ($k = 12$)

For $s = j \in \{1, 2, \dots, 11\}$ at $k = 12$, the kernel $\tilde{k}_T(\tau, j)$ collapses to a polynomial in τ via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$. Three representative values (matching the main note's worked example):

$$\tilde{k}_T(\tau, 1) = -\tau - \frac{1}{2} + \frac{1}{11} B_{11}(\tau + 1), \quad (22)$$

$$\tilde{k}_T(\tau, 6) = -\frac{1}{3} B_6(\tau + 1), \quad (23)$$

$$\tilde{k}_T(\tau, 11) = -\frac{1}{11} B_{11}(\tau + 1) + \tau + \frac{1}{2}. \quad (24)$$

At these integer $s = j$, the orbit sum (13) becomes a sum of polynomial-in- $\gamma_0 \tau_p$ values, each weighted by $(c\tau_p + d)^{10} \sigma^T(\gamma_0)$. For example, the identity-orbit-point contribution at $s = 6$ is

$$\mathcal{R}_{\tau_p}(f, 6)|_{\tau_0=i} = -2\pi i i^{-6} \sigma^T(I) \text{Res}(f, \tau_p) \left(-\frac{1}{3} B_6(\tau_p + 1)\right) = \frac{2\pi i}{3} \sigma^T(I) \text{Res}(f, \tau_p) B_6(\tau_p + 1) \quad (25)$$

using $i^{-6} = -1$. The T -orbit point at $s = 6$:

$$\mathcal{R}_{\tau_p+1}(f, 6)|_{\tau_0=i} = \frac{2\pi i}{3} \sigma^T(T) \text{Res}(f, \tau_p) B_6(\tau_p + 2). \quad (26)$$

The S -orbit point at $s = 6$:

$$\mathcal{R}_{-1/\tau_p}(f, 6)|_{\tau_0=i} = \frac{2\pi i}{3} \sigma^T(S) \operatorname{Res}(f, \tau_p) \tau_p^{10} B_6(-1/\tau_p + 1). \quad (27)$$

And so on. In particular, the integer- s specialisations are $\mathbb{Q}$ -polynomial in the orbit-point $\gamma_0 \tau_p$ (times the cocycle factor and $2\pi i/3$), so the running total (21) at integer s is a $\mathbb{Q}$ -rational linear combination of $\operatorname{Res}(f, \tau_p)$, evaluated against an explicit polynomial in the orbit data $(\gamma_0 \tau_p, (c\tau_p + d)^{10}, \sigma^T(\gamma_0))$.

Stripping further to $s = 2$, $k = 12$: the explicit Poincaré sum

Specialising the kernel at $s = 2$, $k = 12$:

$$\tilde{k}_T(\tau, 2) = \zeta(-1, \tau + 1) - e^{i\pi} \zeta(-9, \tau + 1) = -\frac{1}{2} B_2(\tau + 1) - \frac{1}{10} B_{10}(\tau + 1), \quad (28)$$

using $e^{i\pi} = -1$ and $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$. The two Bernoulli polynomials, written in shifted form $B_m(\tau + 1)$:

$$B_2(\tau + 1) = \tau^2 + \tau + \frac{1}{6}, \quad B_{10}(\tau + 1) = \tau^{10} + 5\tau^9 + \cdots + \frac{5}{66} \quad (29)$$

(a degree-10 polynomial in τ with explicit rational coefficients). Substituting (28) into (13) gives the orbit sum at $s = 2$ as a sum of two explicit Bernoulli-Poincaré sums:

$$K_f^T(\tau_p, 2) = -\frac{1}{2} \sum_{\gamma_0 \in \Gamma_{\tau_p} \setminus \Gamma} (c\tau_p + d)^{10} \sigma^T(\gamma_0) B_2(\gamma_0 \tau_p + 1) - \frac{1}{10} \sum_{\gamma_0} (c\tau_p + d)^{10} \sigma^T(\gamma_0) B_{10}(\gamma_0 \tau_p + 1). \quad (30)$$

This is the analog of (13) the user asked for: two explicit orbit sums, each a Poincaré average of a Bernoulli polynomial along $\Gamma \cdot \tau_p$, weighted by the cocycle factor $(c\tau_p + d)^{10}$ and the winding number $\sigma^T(\gamma_0)$.

Expanding the cocycle-weighted Bernoulli into moment Poincaré sums. Using $\gamma_0 \tau_p + 1 = ((a + c)\tau_p + (b + d))/(c\tau_p + d)$ and $B_2(z) = z^2 - z + \frac{1}{6}$, a short computation gives $B_2(\gamma_0 \tau_p + 1) = (\gamma_0 \tau_p)^2 + \gamma_0 \tau_p + \frac{1}{6}$, and so

$$(c\tau_p + d)^{10} B_2(\gamma_0 \tau_p + 1) = (a\tau_p + b)^2 (c\tau_p + d)^8 + (a\tau_p + b)(c\tau_p + d)^9 + \frac{1}{6} (c\tau_p + d)^{10}. \quad (31)$$

Each term on the right is a degree-10 polynomial in τ_p of the form $(a\tau_p + b)^j (c\tau_p + d)^{10-j}$ for $j \in \{0, 1, 2\}$. Similarly, $(c\tau_p + d)^{10} B_{10}(\gamma_0 \tau_p + 1)$ expands as a $\mathbb{Q}$ -linear combination of $(a\tau_p + b)^j (c\tau_p + d)^{10-j}$ for $j \in \{0, 1, \dots, 10\}$ — all eleven moments contribute.

Define the moment Poincaré sums

$$P_j(\tau_p) := \sum_{\gamma_0 \in \Gamma_{\tau_p} \setminus \Gamma} \sigma^T(\gamma_0) (a\tau_p + b)^j (c\tau_p + d)^{10-j}, \quad j = 0, 1, \dots, 10. \quad (32)$$

Then (30) rearranges into a finite $\mathbb{Q}$ -linear combination

$$K_f^T(\tau_p, 2) = \sum_{j=0}^{10} q_j P_j(\tau_p), \quad (33)$$

with explicit rational coefficients q_j collecting the contributions of B_2 and B_{10} at each moment-order j . (The B_2 part contributes only to $j \in \{0, 1, 2\}$; the B_{10} part contributes to all j .)

4.2 Period-integral at $\tau_0 = i$: orbit sum on V_{k-2}

The cocycle residue jump (main note, Corollary on the period-polynomial wall-crossing residue) reduces at $\tau_0 = i$ to the T -only contribution

$$\Delta C_T^f(\tau_0 = i) = 2\pi i (2\pi i)^{k-2} \sigma_{\tau_p}^T \text{Res}(f, \tau_p) (X - \tau_p Y)^{k-2}. \quad (34)$$

Summed over the orbit:

$$\Delta C_T^f|_{\text{orbit}} = 2\pi i (2\pi i)^{k-2} \text{Res}(f, \tau_p) \sum_{\gamma_0} \sigma^T(\gamma_0) (c\tau_p + d)^{k-2} (X - \gamma_0 \tau_p Y)^{k-2}. \quad (35)$$

The product $(c\tau_p + d)^{k-2} (X - \gamma_0 \tau_p Y)^{k-2}$ simplifies via the classical identity

$$(c\tau_p + d)(X - \gamma_0 \tau_p Y) = (dX - bY) - \tau_p (aY - cX) = X' - \tau_p Y', \quad (36)$$

Here $X' = dX - bY = X|_{\gamma_0^{-1}}$ and $Y' = -cX + aY = Y|_{\gamma_0^{-1}}$ are the images of the auxiliary variables under the V_{k-2} -slash action of $\gamma_0^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$: by the standard convention $P|_{\gamma}(X, Y) := P(aX + bY, cX + dY)$ applied to the inverse matrix γ_0^{-1} . Raising (36) to the $(k-2)$ -th power:

$$(c\tau_p + d)^{k-2} (X - \gamma_0 \tau_p Y)^{k-2} = (X' - \tau_p Y')^{k-2}. \quad (37)$$

The right side, regarded as a polynomial in the original auxiliary variables (X, Y) , is the slash translate of the seed polynomial $(X - \tau_p Y)^{k-2}$: substituting $X \rightarrow X', Y \rightarrow Y'$ inside the polynomial $X - \tau_p Y$ literally produces $X' - \tau_p Y'$, and so

$$(X' - \tau_p Y')^{k-2} = (X|_{\gamma_0^{-1}} - \tau_p Y|_{\gamma_0^{-1}})^{k-2} = ((X - \tau_p Y)^{k-2})|_{\gamma_0^{-1}} \quad (38)$$

by the definition of the slash action. Combining (37) and (38):

$$(c\tau_p + d)^{k-2} (X - \gamma_0 \tau_p Y)^{k-2} = ((X - \tau_p Y)^{k-2})|_{\gamma_0^{-1}}, \quad (39)$$

the γ_0^{-1} -slash translate of the Eichler polynomial kernel $(X - \tau_p Y)^{k-2}$. Substituting into (35):

$$\Delta C_T^f|_{\text{orbit}} = 2\pi i (2\pi i)^{k-2} \text{Res}(f, \tau_p) \sum_{\gamma_0 \in \Gamma_{\tau_p} \backslash \Gamma} \sigma^T(\gamma_0) ((X - \tau_p Y)^{k-2})|_{\gamma_0^{-1}}. \quad (40)$$

This is a Poincaré averaging on the polynomial coefficient system V_{k-2} , with the seed polynomial $(X - \tau_p Y)^{k-2}$ (the standard Eichler kernel at the pole τ_p) translated by the γ_0^{-1} -action and weighted by the T -segment winding-numbers $\sigma^T(\gamma_0)$. No analytic kernel is involved at the level of V_{k-2} — the orbit sum is a purely V_{k-2} -valued Poincaré sum.

Remark 4.1. (40) should be compared directly with HR1919's Theorem 2 in the form $f(q) = 2\pi i A \sum (a + b\alpha)^{-(n+2)} \mathbf{q}^2 / (q^2 - \mathbf{q}^2)$: the orbit-of-residues structure is identical, with the auxiliary polynomial variable (X, Y) playing the role of HR's q -variable, and the slash action $|_{\gamma_0^{-1}}$ on V_{k-2} playing the role of HR's translated $\mathbf{q}$ -kernel. The integer- $s = j$ specialisation of (12) is the coefficient of $\binom{k-2}{j} (-1)^j X^{k-2-j} Y^j$ in (40), providing the dictionary between the L -integral and period-polynomial sides of the orbit-sum construction at $\tau_0 = i$.

4.3 The $k < 0$ regime: HR1919 convergence

For $k < 2$, in particular $k < 0$, the cocycle factor $(c\tau_p + d)^{k-2}$ has exponent $k - 2 < -2$ and *decays* in $|c\tau_p + d|$ as $|c|, |d| \rightarrow \infty$. Standard Petersson-Poincaré-series convergence then applies:

$$\sum_{(c,d)=1} |c\tau_p + d|^{k-2} < \infty \quad \text{for } k - 2 < -2, \text{ i.e., } k < 0. \quad (41)$$

Combined with the boundedness of $\tilde{k}_T(\gamma_0\tau_p, s)$ on the orbit for $\text{Re}(s)$ in suitable ranges (using $|\zeta(\sigma, a)| = O(|\text{Im } a|^{1/2 - \text{Re } \sigma + \varepsilon})$ on vertical lines), the orbit sum $K_f^T(\tau_p, s)$ converges absolutely for $k < 0$ and $\text{Re}(s)$ in an appropriate strip.

This is the HR1919 regime: they study modular functions of small (typically negative-in-our-convention) weight n , where the Poincaré-orbit-of-residues sum converges directly. In our convention with weight k :

- $k < 0$: orbit sums converge naively (HR1919-style).
- $k = 0$: orbit sums marginally divergent / require regularisation.
- $k > 2$: orbit sums require regularisation; the main note's $k = 12$ falls here.

DJ2008 duality and dual-weight convergence. DJ2008 Corollary 1 ([2, Cor. 1]) establishes the coefficient duality $a_k(m, n) = -a_{2-k}(n, m)$ between weakly holomorphic forms of weights k and $2 - k$. In our context: the dual of weight $k = 12$ is weight $2 - k = -10$, which sits in the $k < 0$ regime where the orbit-sum convergence is naive/automatic.

This suggests a clean route to making sense of the cumulative wall-crossing $K_f^T(\tau_p, s)$ at $k = 12$:

1. Establish the orbit sum at the dual weight $k = -10$, where convergence is automatic.
2. Use the DJ2008 duality to transport the orbit-sum identification back to weight $k = 12$.
3. The duality maps the principal-part data ($\text{Res}(f, \tau_p)$) at $k = 12$ to the corresponding V_{k-2} -coefficient data at weight -10 , where the Poincaré-orbit sum directly converges.

Remark 4.2. *The polynomial coefficient system V_{k-2} has dimension $k - 1$ for $k \geq 2$ and is empty (in the polynomial sense) for $k < 2$. For $k < 0$, the "polynomial" $(X - \tau Y)^{k-2}$ is a formal rational function / Laurent expansion in (X, Y) rather than a polynomial — so the period-polynomial framework of the main note (Theorem 1.1) does not directly transport to $k < 0$ without reinterpretation. Nevertheless the L -integral side (13) is well-defined for any $k \in \mathbb{Z}$ via the analytic kernel $\tilde{k}_T(\tau, s)$, which has no polynomial restriction. The DJ2008 duality applies precisely at the level of the L -integral / coefficient data, not directly at the polynomial V_{k-2} level.*

Concrete test case at $k = -10$. Take $f = 1/\Delta(\tau) \in M_{-12}^!$ — a weakly holomorphic weight- (-12) form with a q^{-1} pole at the cusp and no interior poles. To get an interior-pole example at weight -10 , take $f = E_4(\tau)/\Delta(\tau) \in M_{-8}^!$... still no interior poles. A worked example with interior poles at weight -10 would need a form like $E_4^3 E_6 / (\Delta^2(j - j(\alpha)))$ for α a CM point; this has weight $4 \cdot 3 + 6 - 12 \cdot 2 = -6$. Picking weights more carefully to land at -10 with an interior simple pole at α is a concrete numerical experiment that would test the naive-convergence claim of (13) at $k < 0$.

5 The standard-Mellin contour: cumulative residue at arbitrary k

This section addresses the question: what does the cumulative residue sum look like in the simplest possible setting — where the integration contour can be pushed back to the classical Mellin contour $\tau \in [0, i\infty]$, the integrand simplifies to $f(\tau) \tau^{s-1}$, and the modular form f is taken to vanish at both cusps so that the contour endpoints are not problematic? The question is: what *modular form* (in τ_p) does the orbit sum of residues equal?

We work at arbitrary even weight $k \in \mathbb{Z}$ throughout, assuming that the chosen meromorphic f has been arranged (by dividing by enough powers of $1/j(\tau)$, or equivalent) to vanish at both cusps. This sidesteps convergence questions at the cusps; we make no claim about convergence of the resulting orbit sum, treating it purely as a formal algebraic object.

5.1 Single-crossing residue at the Mellin contour

For f meromorphic of weight k vanishing at both cusps, the classical Mellin integrand $\tau^{s-1} f(\tau)$ is integrable over $[0, i\infty]$ except where the contour passes through interior poles of f on the imaginary axis or in the swept region. A single crossing of an interior pole τ_p of f contributes

$$\mathcal{R}_{\tau_p}^{\text{Mellin}}(f, s) = 2\pi i \sigma_{\tau_p} \tau_p^{s-1} \text{Res}(f, \tau_p), \quad (42)$$

with $\sigma_{\tau_p} \in \{-1, +1\}$ the orientation of the crossing (positive for a deformation that newly encloses τ_p). The integrand $\tau^{s-1} f(\tau)$ has a simple pole at τ_p with residue $\tau_p^{s-1} \text{Res}(f, \tau_p)$ by elementary calculus.

5.2 Cumulative orbit-sum residue

Suppose f has interior poles at a single Γ -orbit $\Gamma \cdot \tau_p$ (generic τ_p , so $\Gamma_{\tau_p} = \{\pm I\}$). Deform the contour continuously to enclose every point in the orbit exactly once with the same orientation $\sigma \equiv +1$. Using the residue-transport identity (3) and (42), the cumulative residue across the full orbit is

$$\Delta L^*(f, s)|_{\text{full orbit}} = 2\pi i \sum_{\gamma_0 \in \Gamma/\{\pm I\}} (\gamma_0 \tau_p)^{s-1} (c\tau_p + d)^{k-2} \text{Res}(f, \tau_p). \quad (43)$$

Substituting $(\gamma_0 \tau_p)^{s-1} = (a\tau_p + b)^{s-1} / (c\tau_p + d)^{s-1}$:

$$\Delta L^*(f, s)|_{\text{full orbit}} = 2\pi i \text{Res}(f, \tau_p) \mathcal{P}_k(\tau_p, s), \quad (44)$$

where the orbit sum reduces to

$$\mathcal{P}_k(\tau_p, s) := \sum_{\gamma_0 \in \Gamma/\{\pm I\}} (a\tau_p + b)^{s-1} (c\tau_p + d)^{k-1-s}. \quad (45)$$

This is the explicit Poincaré-style sum the user asked for: every summand is a product of two simple linear factors in τ_p , with exponents that sum to $k-2$ and that split between the upper-row entries (a, b) (exponent $s-1$) and the lower-row entries (c, d) (exponent $k-1-s$) of γ_0 . Nothing else floats: no Hurwitz zetas, no Bernoulli polynomials, no winding numbers beyond the uniform $\sigma \equiv +1$ choice.

5.3 Structural properties of $\mathcal{P}_k(\tau_p, s)$

Weight in τ_p . Under $\tau_p \mapsto \gamma_1 \tau_p$ for $\gamma_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \in \Gamma$, the summand transforms as

$$(a(\gamma_1 \tau_p) + b)^{s-1} (c(\gamma_1 \tau_p) + d)^{k-1-s} = \frac{(\tilde{a}\tau_p + \tilde{b})^{s-1} (\tilde{c}\tau_p + \tilde{d})^{k-1-s}}{(c_1 \tau_p + d_1)^{(s-1)+(k-1-s)}}, \quad (46)$$

where $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d})$ are the entries of γ_1 . The exponent sum is $(s-1) + (k-1-s) = k-2$, so the denominator in (46) is $(c_1 \tau_p + d_1)^{k-2}$, independent of γ . Reindexing $\gamma \rightarrow \gamma \gamma_1$ in the sum:

$$\mathcal{P}_k(\gamma_1 \tau_p, s) = (c_1 \tau_p + d_1)^{-(k-2)} \mathcal{P}_k(\tau_p, s) = (c_1 \tau_p + d_1)^{2-k} \mathcal{P}_k(\tau_p, s). \quad (47)$$

So $\mathcal{P}_k(\tau_p, s)$ is a weight- $(2-k)$ modular object in τ_p for every s . This is the structural answer: the cumulative orbit-residue has the modular weight *dual* to the weight of f . At $k = 12$ (our running case) it is weight -10 in τ_p ; at $k = 0$ it is weight $+2$ in τ_p .

Functional equation $s \leftrightarrow k-s$. The summand $(a\tau_p + b)^{s-1}(c\tau_p + d)^{k-1-s}$ is invariant under the simultaneous swap $s \leftrightarrow k-s$ and $(a, b) \leftrightarrow (c, d)$. The involution $\gamma \mapsto \tilde{\gamma} = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix}$ preserves $\det = 1$ and swaps the roles of the upper and lower rows (up to the sign on the new upper row). Pairing each γ with its $\tilde{\gamma}$ gives the identity

$$\mathcal{P}_k(\tau_p, k-s) = (-1)^{k-s-1} \mathcal{P}_k(\tau_p, s), \quad (48)$$

formally. For even k and integer s , this is the classical $s \leftrightarrow k-s$ symmetry of an L -function, transmitted to the orbit sum.

Total τ_p -degree per summand. Each summand $(a\tau_p + b)^{s-1}(c\tau_p + d)^{k-1-s}$, formally expanded in τ_p , has total polynomial degree $k-2$ (when s is a non-negative integer with $0 \leq s-1 \leq k-2$). The summand is homogeneous of degree $k-2$ in the τ_p -direction.

Integer- s specialisations. At $s = 1$: the upper-row power vanishes, leaving $\mathcal{P}_k(\tau_p, 1) = \sum (c\tau_p + d)^{k-2}$.

At $s = k-1$: the lower-row power vanishes, leaving $\mathcal{P}_k(\tau_p, k-1) = \sum (a\tau_p + b)^{k-2}$.

For $s \in \{2, \dots, k-2\}$ in the interior of the critical strip, both factors are nontrivial, with the upper-row exponent $s-1$ and lower-row exponent $k-1-s$ both positive integers summing to $k-2$.

5.4 What kind of modular form is $\mathcal{P}_k(\tau_p, s)$?

The orbit sum $\mathcal{P}_k(\tau_p, s)$ is formally a weight- $(2-k)$ modular form in τ_p . The natural candidate identifications:

- **Eisenstein-style series at weight $2-k$.** The classical holomorphic Eisenstein series E_{2-k} exists only for $2-k \geq 4$, i.e. $k \leq -2$. For $k > 2$, E_{2-k} is not holomorphic-modular (negative weight has no holomorphic forms on $\mathrm{SL}_2(\mathbb{Z})$); but weakly holomorphic / meromorphic modular forms of weight $2-k$ do exist (e.g. $1/\Delta$ at $k = 12$ is weight -12 , and combinations with j give weight -10). Whether $\mathcal{P}_k$ identifies with a specific weakly holomorphic weight- $(2-k)$ form is an open question.
- **Γ -orbit sums of polynomial moments.** Expanding $\mathcal{P}_k(\tau_p, s)$ at integer $s = j \in \{1, \dots, k-1\}$ gives the moment Poincaré sums $\sum (a\tau_p + b)^{j-1} (c\tau_p + d)^{k-1-j}$, indexed by j , each of which is a weight- $(2-k)$ object in τ_p .

- **Lewis–Zagier period functions at weight $2 - k$.** The Lewis–Zagier theory [?] constructs period functions for Maass wave forms parameterised by s , with weight-related transformation in a complementary variable. $\mathcal{P}_k(\tau_p, s)$ has the right shape (weight $2 - k$ in τ_p , analytic in s , orbit-sum structure) to be a candidate period-function object in the Lewis–Zagier family.
- **DJ2008 duality at weight $2 - k$.** Duke–Jenkins [2] establish the duality $a_k(m, n) = -a_{2-k}(n, m)$ between weakly holomorphic forms of weights k and $2 - k$. Our orbit sum being weight- $(2 - k)$ — exactly the DJ2008-dual weight — is not a coincidence. $\mathcal{P}_k(\tau_p, s)$ is the natural candidate for a one-parameter family in s living in $M_{2-k}^!$ (or its s -extension), parameterised by the pole-location τ_p of the weight- k form f .

Honest open question. Is there a closed-form identification

$$\mathcal{P}_k(\tau_p, s) \stackrel{?}{=} \sum_m c_m(s) f_{2-k, m}^!(\tau_p),$$

where $\{f_{2-k, m}^!\}$ is the DJ2008 basis of weakly holomorphic weight- $(2 - k)$ forms and $\{c_m(s)\}$ are explicit (rational in s , or analytic) coefficients? At integer s , the right side would be a finite sum. At general s , an analytic expansion in the basis. This is the cleanest possible form of “the cumulative wall-crossing is an interesting modular form” — and it is the question worth testing with a worked example, but we set numerics aside per the present scope.

6 The conjectural identification

For f with poles at $\Gamma \cdot \tau_p$ and our cumulative wall-crossing residue $K_f(\tau_p, s)$, two cases stand out.

6.1 Integer $s = j \in \{2, \dots, k - 2\}$

At integer $s = j$ in the critical strip, $\tilde{k}_T(\tau, j)$ collapses to a polynomial in τ via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$. The orbit sum

$$K_f(\tau_p, j) = \sum_{\gamma_0} (c\tau_p + d)^{k-2} \left[\sigma^S(\gamma_0) (\gamma_0 \tau_p)^{j-1} + \sigma^T(\gamma_0) \tilde{k}_T(\gamma_0 \tau_p, j) \right] \quad (49)$$

is, formally, the orbit sum of a weight- k modular form valued at τ_p , with the explicit polynomial kernels evaluated along the orbit.

Conjecture 6.1. For $\ell = 0$ (i.e. $k \in \{4, 6, 8, 10, 14\}$, where $\dim S_k = 0$), the integer- $s = j$ specialisation $K_f(\tau_p, j)$ admits an explicit expansion in the DJ2008 basis $\{f_{k, m}(\tau_p)\}_{m \geq 0}$, with coefficients that depend on j and on the Laurent data of f at τ_p . For $\ell \geq 1$ (i.e. $k \geq 12$, $\dim S_k \geq 1$), $K_f(\tau_p, j)$ similarly expands in $\{f_{k, m}\}_{m \geq -\ell}$, with the $m \leq 0$ part contributing the residue / quasi-period data.

This would identify our integer- s wall-crossing residue family with a known basis-expansion structure in $\mathcal{M}_k$. The natural proof route would be to set up the DJ2008 generating-function identity (9) with $z = \tau_p$ and our kernel $\tilde{k}_T(\tau, j)$ playing the role of $f_{2-k}(\tau)$ (suitably modified to account for the segment-specific contributions σ^S, σ^T).

6.2 Continuous $s \in \mathbb{C}$

At continuous s , $\tilde{k}_T(\tau, s)$ is not a polynomial in τ ; it is the analytic family of Hurwitz-zeta combinations. The orbit sum $K_f(\tau_p, s)$ is, formally, a one-parameter analytic family of weight- k meromorphic modular forms on $\mathbb{H} \setminus \Gamma \cdot \tau_p$, parameterised by $s \in \mathbb{C}$. Two questions:

1. **Convergence.** Naive convergence of (5) requires the kernel decay along $\Gamma \cdot \tau_p$ to beat the cocycle growth $|c\tau_p + d|^{k-2}$. Standard Hurwitz-zeta bounds $|\zeta(\sigma, a)| = O(|\operatorname{Im} a|^{1/2 - \operatorname{Re} \sigma + \varepsilon})$ on vertical lines suggest absolute convergence is delicate and may require Res in particular ranges. Regularisation along Niebur–Selberg lines may be required.
2. **Identification as a canonical object.** If $K_f(\tau_p, s)$ converges (or admits canonical regularisation), it is a weight- k meromorphic modular form on $\mathbb{H} \setminus \Gamma \cdot \tau_p$ analytic in s . The natural identification target is a Lewis–Zagier-style s -extension of the DJ2008 basis, or a Niebur–Maass Poincaré series of weight k with s -parameter, or some closely related object.

7 Open questions / next steps

1. Prove or disprove Conjecture 6.1 at integer s for a worked $k = 12$ example: take f a specific meromorphic weight-12 form with a single Γ -orbit of simple poles (e.g. $E_4^3/(j - j(\alpha))$ for α a CM point), and check whether $K_f(\tau_p, j)$ at $j \in \{2, \dots, 10\}$ expands as a finite linear combination of $\{f_{12,m}\}_{m \geq -1} = \{\Delta, \hat{\Delta}, f_{12,2}, \dots\}$.
2. Establish convergence of (5) in some Res-strip. The growth/decay balance: $|c\tau_p + d|^{k-2}$ growth from the cocycle vs. $|\zeta(1 - s, \gamma_0\tau_p + 1)| \cdot |\zeta(s - (k - 1), \gamma_0\tau_p + 1)|$ kernel growth along orbit-points approaching the real axis. This is a standard Petersson-style convergence question.
3. For non-convergent Res, identify the canonical regularisation (analytic continuation in s , Niebur-style regulator, Petersson Poincaré-series limit, etc.).
4. Compare with the polylog projection $\hat{P}_{\tau_p}$ of [3, §2.2]. The polylog projection subtracts the polar tails at τ_p ; our $K_f(\tau_p, s)$ adds up the residue contributions across the orbit. Are these dual constructions? Specifically, is $K_f(\tau_p, s)$ at integer s equal (up to known factors) to the polylog-projected residual?
5. Connect with the Brown–Hain iterated-Eichler-integral framework on $\mathcal{M}_{1,1}$ (cf. [4]): the s -parameterised orbit sum $K_f(\tau_p, s)$ should fit into the iterated-integral / mixed-Hodge story for meromorphic forms with interior poles.
6. For $\ell = 0$ specifically, check whether $K_f(\tau_p, j)$ at $j \in \{2, \dots, k - 2\}$ has a direct expression via DJ2008’s generating function (9) specialised at $\tau = \tau_p$ — this would be the cleanest possible match.

8 Summary

The substantive output of this companion note, after the iterative exploration recorded above:

1. **Structural slot: HR1919 / DJ2008, not classical Petersson Poincaré series.** The cumulative wall-crossing residue $K_f(\tau_p, s)$ of the finite-contour L -integral lives in the orbit-of-residues lane (HR1919 Theorem 2, DJ2008 Theorem 2), *not* the classical convergent-

Petersson-Poincaré-series lane. DJ2008 themselves explicitly avoid Petersson series for $\ell \neq 0$ and use residue-integral reconstruction instead; that is exactly our lane.

2. **The clean algebraic answer (§5).** For meromorphic f of arbitrary even weight k vanishing at both cusps, with contour pushed back to the classical Mellin path $[0, i\infty]$, the cumulative orbit-residue of the L -integral is, formally,

$$\Delta L^*(f, s)|_{\text{full orbit}} = 2\pi i \operatorname{Res}(f, \tau_p) \mathcal{P}_k(\tau_p, s), \quad \mathcal{P}_k(\tau_p, s) = \sum_{\gamma_0 \in \Gamma/\{\pm I\}} (a\tau_p + b)^{s-1} (c\tau_p + d)^{k-1-s}.$$

This is a clean Poincaré-style orbit sum with only the modular cocycle data of γ_0 and the parameter s floating.

3. **Weight $2 - k$ in τ_p , not weight k .** The cumulative orbit-residue $\mathcal{P}_k(\tau_p, s)$ transforms with modular weight $2 - k$ under $\tau_p \mapsto \gamma_1 \tau_p$ — the *dual* weight to the weight of f . At $k = 12$, this is weight -10 , which lives in $M_{-10}^!$ (DJ2008’s dual basis to $M_{12}^!$). This is the central structural observation of this companion note.
4. **Classical $s \leftrightarrow k - s$ functional symmetry.** $\mathcal{P}_k(\tau_p, k - s) = (-1)^{k-s-1} \mathcal{P}_k(\tau_p, s)$ follows from a row-swap involution on Γ , inheriting the classical L -function functional equation at the level of the cumulative-residue object.
5. **The honest open question.** Does $\mathcal{P}_k(\tau_p, s)$ admit an explicit identification $\mathcal{P}_k(\tau_p, s) = \sum_m c_m(s) f_{2-k, m}^!(\tau_p)$ in the DJ2008 basis at the dual weight $2 - k$, with $\{c_m(s)\}$ explicit (rational in s , or analytic)? This is the cleanest possible form of “the cumulative wall-crossing equals an interesting modular form.” Settling it numerically at a worked $k = 12$ example with one Γ -orbit of interior poles (e.g. $f(\tau) = E_4^3/[\Delta(j(\tau) - j(\tau_p))]$ at a CM τ_p) is the natural next step. Set aside per the present scope.
6. **Eleven moment Poincaré sums at integer $s, k = 12$.** At $\tau_0 = i$, $s = 2$, $k = 12$, the cumulative residue expands as a $\mathbb{Q}$ -linear combination of the moment Poincaré sums $P_j(\tau_p) = \sum (a\tau_p + b)^j (c\tau_p + d)^{10-j}$, $j = 0, \dots, 10$ (§4.1). These moments are the V_{10} -monomial coefficients of the period-polynomial orbit sum; the L -integral wall-crossing at integer s and the period polynomial wall-crossing are the same eleven moments read off through different kernels. (This was previously written up as a “structural payoff” claim above; the user correctly identified it as a trivial restatement of the main-note Thm 1.1 $\leftrightarrow$ Thm 1.2 lockstep, not a genuine structural surprise; that paragraph is preserved below `\end{document}` for the record.)
7. **Convergence regime: $k < 0$ in our convention.** The cocycle factor $(c\tau_p + d)^{k-2}$ decays for $k - 2 < -2$, i.e. $k < 0$. The HR1919 setting; our main-note $k = 12$ case falls outside this naive-convergence regime and requires either truncation to finite swept-region or canonical regularisation. The DJ2008 weight-duality $k \leftrightarrow 2 - k$ is the natural route: identify $\mathcal{P}_k$ at $k = 12$ with a canonical object at weight -10 where the orbit sums are naively convergent.

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